

1 Introduction

Cross-document coreference resolution (CDCR) is an NLP task to identify coreferential mentions between discourse elements, such as entities, events, or more complex semantic discourse elements (Des). The existing methodologies for annotation of CDCR datasets include event-centric (ECB+) and concept-centric annotation schemes (NewsWCL50). The former may result in too narrow and the latter in too broad coreferential chains. Thus, we propose an annotation methodology that identifies narrow DE-centric coreference chains. We aim to identify complete coreferential chains. The following coding book uses the terms “coreference chain” and “discourse element” interchangeably.

The goal of this study is to improve the annotation scheme of NewsWCL50 to identify DEs that:

- 1) are covered across a set of related articles (which mentions occur in multiple articles)
- 2) describe a set of articles (that are specific enough to report about a topic/event), and that
- 3) contain as many and as long as possible DE mentions to represent the lexical diversity of the mentions.

Therefore, we conduct a manual deductive content analysis. This codebook provides annotation rules for finding coreference mentions, e.g., based on descriptions of possible concept types, and for linking concept mentions. We provide examples of annotations below.

When referring to a semantic concept, such as an actor in a news event, authors choose from multiple words or multi-word terms, including Merkel, Angela Merkel, or Chancellor Angela Merkel. While these three exemplary terms are considered neutral in this study, other terms may not be. If a chosen term may induce a different (emotional) judgment on the referred concept by a human reader, the term used by the author induces bias by word choice. Examples of bias by word choice include regime vs. government, dictator vs. chief of state, Kim Jong Un vs. Rocket Man, and coronavirus vs. boomer-remover.

Likewise, authors can label the same concept differently; i.e., adding different attributes to the concept leads to a different perception of the concept by the human reader in the text. If a chosen attribute induces a different (emotional) judgment of its concept by a human reader, the attribute causes bias by labeling. Examples of bias by labeling include coalition forces vs. invasion forces, immigrants vs. undocumented immigrants (where the former does not contain any label), and an anxious and uncertain President Trump.

Coreferential chains often consist of strict identity relations, but border cases, such as freedom fighters - terrorists, young undocumented immigrants - aliens, yield less strict or so-called near-identity relations. Such identity relations are often context-specific or are a product of a public’s figure statement that introduces identity between two mentions, e.g., “Trump called Kim Jong Un a Rocket Man because Kim didn’t want to give up his nuclear weapons.”

Unlike the event-based methodology, the concept-centric methodology aims to identify mentions of more semantically complex concepts and aggregate them. When reading news articles, some specific actions or events may be mentioned only a few times and thus be considered infrequent. However, such concepts typically require an overview from a more general perspective and form a concept when considered as pieces of a jigsaw puzzle. For example, the concept “Migration to the US” encompasses a chain of subevents detailing how immigrants marched to the border, as well as a generalized mention of this process as “migration.”

The coding book contains general definitions (S2) and annotation rules (S3).

2 Definitions and Examples

2.1 General remarks

All examples, links, and referred resources provided in this document were valid at the time of writing, but might not necessarily be when reading.

2.2 Term

We define a term as a single word or a multi-word expression in a text.

2.3 Concept

We define a concept in the linguistic sense as a single, language-independent, semantic unit. In a document, a concept is represented by a term. Synonyms, for instance, represent the same concept (or similar concepts). A concept, for instance, is the current President of the United States, which might be represented in a document also as “Trump,” “President Trump,” “the President,” “Donald Trump,” and so on. In this study, concepts are usually:

- 1) entities, e.g., politicians, geo-political entities (GPEs), locations, objects,
- 2) events, such as actions “deport” or “overstep”, or more general more as a process such as “immigration”
- 3) broader and aggregating concepts such as reaction to some events, consequences of an event, and a chain of interrelated events.

2.4 Identity relations

- Clear identity between two mentions that refer to one concept, e.g., Donald Trump – the President.
- Can be found between NPs and VPs, e.g., “met with Donald Trump” – “the planned meeting with Donald”
- Identity relations are to be found among the mentions with extensive lexical diversity, i.e., caused by words added to a concept, also called attributes, or by synonyms. An **attribute** is typically an adjective (modifying subjects or objects) or an adverb (modifying predicates), but can also be a noun as a noun modifier or a compound. Examples: an anxious and uncertain President Trump; a difficult, emotional decision for the president; less-educated, native-born Americans; a cruel action. Synonyms: talked about – discussed, DACA recipients – undocumented children, Olympics – sport competition.

2.5 Near-identity relations

- More sophisticated identity relations due to a part of the whole relations (metonym and/or meronym) between mention and a concept, while still referring to the whole concept, e.g., the Kremlin = the Russian government.
- Words that originally have different meanings but in a specific context refer to the same concept while inducing judgment in the reader, e.g., invade vs. cross border.
- Relations established by so-called corporal verbs, i.e., be, smell, feel, etc., e.g., “*This meeting is a big step forward.*” Although these mentions do not have strict identity, such a phrase introduces a new associated description to the concept “meeting”. We expand the list of corporal verbs with “calling” verbs, i.e., call, name, describe, denounce, etc., e.g., “Trump called *Kim Jong Un* a *Rocket Man*.” We also annotate mentions, one of which is a noun and the second is a dangling attribute, e.g., “Khan said

that *Trump* behaved like *a 12-year-old*". Sometimes concepts are associated with a mention through a close context between the mentions, but lack a connecting word, yet such mentions need to be annotated as coreferential.

2.6 Aggregating relations

- An article/a set of articles can contain multiple related small concepts (up to six mentions) that do not have identity relations, but these concepts are clearly related to each other within a text context.
- For example, a concept "Interactions with Trump" includes mentions referring to dinners, meetings, and conversations with Donald Trump, which James Comey covered in his memos when trying to expose Trump.
- Such aggregating concepts can be found between NPs and VPs and are usually harder to identify because they require a higher level of abstraction than concepts with identity or near-identity relations.

3 What to annotate and how

3.1 Definitions

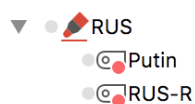
The main units to be coded are concepts.

CONCEPT: A CONCEPT is any semantic concept mentioned in the article, including ACTIONS, ACTORS, ACTOR-Is, ACTOR-MISCs, COUNTRYs, ORGANIZATIONs, EVENTs, GROUPs, MISCs, and OBJECTs. Each CONCEPT's mention can be the target of one or more PROPERTY mentions.

A concept is one of the following types:

ACTION: An action that is performed by an ACTOR (or other CONCEPT) and/or that happens during an EVENT, e.g., the ACTION "Negotiate about the peace" happened during the EVENT "PRK USA Summit." Compared to EVENT, ACTION is typically shorter in time and more narrowly defined. Consists typically mostly of verb phrases (VPs).

ACTOR: A person proper noun that performs an action. It can be any mention of a proper noun in the text. An ACTOR is an individual who is part of or represents a specific group of individuals, e.g., government or a more specific group like "Democrats" or "DACA recipients." An instance of an ACTOR is created with the first mention of a new proper noun and is named with the last name of an actor. In the case of government, an ACTOR concept needs to be created as a child node of the COUNTRY concept (see the figure below), and in the more specific cases as a child node of the GROUP or ACTOR-MISC concepts.



COUNTRY: A geo-political entity (GPE). Use the country's three-letter country code, e.g., United State of America = USA, Russia = RUS, Germany = DEU, see: https://en.wikipedia.org/wiki/ISO_3166-1_alpha-3 Institutions that are part of a country/state (metonyms, typically the government), metonyms, i.e., words that represent a whole entity with mentioning a part of it, e.g., "Moscow" used to address the Russian government, references such as "nation" that indicate a country as a socio-political entity, need to be tagged as the country. For instance, the White House, Pentagon, and the American nation are all permanently associated with or representing the USA, the US Army; therefore, you must code all of them just as USA. Even a single mention of a country is enough to create a code.

EVENT: An event is a continuous action, such as “armed confrontation,” “battle,” “meeting,” “campaign,” etc. An event tends to be represented both with NPs and VPs, sometimes equally.

GROUP: A group of individuals that act collectively, e.g., “immigrants,” “DACA recipients,” or “Trump’s lawyers,” “sports club.” Thus, similar to ACTOR, but is a group of individuals, whereas ACTOR is a single individual. Entities of political parties and their members, e.g., Democrats and Democratic leaders, are to be annotated as GROUPs.

OBJECT: Something that is a non-animated frequently mentioned object, i.e., “Steele dossier,” “DNC servers.” OBJECT can play the role of an active participant, i.e., taking action, or a passive one, e.g., mentioned as a reason for conflict.

ORGANIZATION: An official non-government related organization entity, i.e., “People without Borders” or “WikiLeaks.”

MISC: Any semantic concept that is mentioned frequently in the article and that does not fit into the other categories. Typically, broadly defined or abstract, it requires a higher level of generalization to identify it, e.g., “Interactions with Trump.” Quite often, a MISC concept contains multiple small or singleton concepts that are connected into a process, e.g., “Denuclearization,” or describes one concept with indirect mentions, e.g., describes “Relations” as both with direct mentions “good relations between Trump and Macros” but also with an indirect mention as “embraced warmly.”

ACTOR-G is a DE that represents a group of people associated with a country or organization.

- **[ENTITY]-I:** Created to annotate individuals that actively represent a COUNTRY code, i.e., take decisions or represent a country, e.g., “Administration officials,” “the European allies,” “Korean envoys”, when no specific reference is given, who are these people. [CONCEPT]-I concept can also be applied to ORGANIZATION if there are non-specific mentions of leaders of this organization. If a subset annotated as -I mentions forms a specific and frequent (mentioned at least four times) group of actors, a separate GROUP needs to be created, e.g., “Immigration officials” as a distinct specific sub-concept of USA-I concepts.
- **[ENTITY]-MISC:** As opposed to the active representation, we use [COUNTRYNAME]-MISC to mark passive membership of some people to a COUNTRY or ORGANIZATION concepts, i.e., as population or members. For example, “American people,” “Iranian citizens”, “Kim’s scientists,” “members of the church.” NOTE: Depending on the context, you will need to decide if instances like “Iranians” denote official representatives or the Iranian people, and correspondingly annotate as -I or as -MISC. Mentions of the population, if occurring more than four times, should be grouped separately, e.g., “demonstrators.”

CONCEPT-TREES: list of all CONCEPTs, specified in MAXQDA, e.g., the ACTOR-TREE contains all ACTORS.

3.2 Workflow of coding for a single news article

Note that in the following, we use the term ACTOR for ease of understanding and readability. However, all ACTOR rules and *mentions* also refer to any other CONCEPT.

- (1) Read the article briefly, thereby scanning for mentions of COUNTRY, ACTOR, and ORGANIZATION concepts involved in the event(s). Create a new code for a concept as soon as you see a first mention of a new concept. Annotate all nominal mentions

referring to these concepts as you see them. Include all attributes used with the head phrase, e.g., President of the United States Donald Trump.

- (2) Finds mentions referring to the [CONCEPT]-I and [CONCEPT]-MISC concepts. If there is a mention of a -MISC concept, e.g., “Iranian people”, but there is no country code IRN created yet, create a code “IRN” and under it “IRN-MISC” concept. Leave the country-DE empty, and annotate mentions referring to the MISC concept. The same is valid for -I concepts.
- (3) Scan for the frequent storyline properties, such as ACTIONS, EVENTS, GROUPs, and OBJECTs. Do they have at least four mentions (nominal, proper nouns) in the current news article? If so, create a new CONCEPT in the ACTOR-LIST. To find how often a CONCEPT occurs, search for its headword of the mention in the search field. Do not fully rely on the search due to the possible high lexical diversity, i.e., a large amount of synonyms that are used to refer to this concept. Annotate nominal occurrences in the articles and include any attribute, i.e., an adjective or adverb, close to the actual reference to the actor, e.g., code 'illegal aliens' but not only 'aliens' (for the DACA Dreamers) or 'Alleged Russian Fighters'.
- (4) After you become more familiar with the text, try to identify more abstract concepts that 1) report about smaller aspects of a story, whose mentions are quite infrequent and are not directly coreferential but are related to each other, e.g., describe reactions to something or steps to prevent immigrants from coming to the US. When you find more than four mentions of such an abstract concept, create a separate code for it and annotate the mentions.

3.3 Coding rules

A1: as long as necessary. Never remove the immediate context of a phrase and its attributes, e.g., “the redacted and declassified *memos* -- running 15 pages in total”. If there is an appositional modifier, e.g., “Flynn, *the national security advisor*” or any related clause describing an entity “Trump, *who hesitated to call first*”, annotate such a long mention as a whole. Phrases cannot be coded in isolation but need to be coded according to their actual meaning in the text. For verbs: double check if you only want to code the verb, and annotate the action and the closest to it modifiers as long as they are related to show relation to the concept (have been blessed vs have been blessed with all the opportunities). Example for NP: 'hostility to any compromise with Americans'. Also, pay attention if you want to code a noun that has an adjective before it; usually, the adjective is a label that describes the noun, and as such, shall not be discarded but coded along with the noun. For instance, “suspected chemical weapons attack”: would be perpetrator if it were only a chemical weapons attack, but including the suspected word in front, it needs to be tagged as other-other. If a mention has a dangling attribute, annotate the longest of it.

B1: Consider context. Some mentions may fall out of usual synonymic relations, but the context will make them synonymous, e.g., “kids that have done nothing wrong” – “DACA recipients”. Annotate mentions that are related in the narrow context of the analyzed set of articles.

3.4 Example

The following is an example of the annotated set of related articles in MAXQDA using the described coding book.

 Code System		 Supporters	26
 Central America	2	 Ramirez	3
 GTM	6	 Blanco	2
 GTM-Misc	3	 Torres	3
 SLV	1	 Suarez	1
 SLV-Misc	3	 Lozano	1
 NIC	1	 People without borders	15
 MEX	34	 Mujica	3
 Tijuana	24	 Mensing	2
 Mex-I	0	 Lawyers	16
 Mex-Misc	4	 Phillips	1
 Nieto	0	 Showing Up For Racial J...	2
 USA	87	 Cronk	2
 Obama	1	 Ramos	2
 Democrats	0	 Saravia	1
 Lofgren	1	 Jordahl	1
 Flores	1	 USA-MEX Border	86
 Scott	4	 Efforts of the border control	40
 Republicans	1	 Migrant caravan	192
 Gonzalez	1	 Cazares	2
 Brown	1	 Avila	1
 Thompson	2	 Mendez	2
 Nielsen	5	 Palacio	1
 Sessions	5	 Gomez	3
 McAleenan	1	 Rodriguez 2	2
 USA-Misc	5	 Rodriguez	4
 USA-I	7	 Claros	3
 Immigration officials	24	 Carillo	1
 Trump	25	 Quantanillo	2
 HND	1	 Migration to the USA	85
 HND-Misc	4	 Interview	8
		 Asylum	11
		 Seek asylum	30

Type	Topic	DE / coref. chain	Mentions
ACTION	0	trip to PRK	meeting met with Kim Jong Un secretly met with North Korean leader Kim Jong Un Mr. Pompeo 's trip `` direct talks at very high levels '' communicated directly with Washington reportedly made the secret visit to North Korea direct talks at `` extremely high levels '' the secret trip direct talks at very high levels , extremely high levels

			Mr. Pompeo 's mission dispatched the C.I.A. director Mike Pompeo to North Korea Pompeo 's trip in direct talks with North Korea at `` extremely high levels , " secretly met with Kim Jong-un a successful negotiation with North Korea secretly traveled to North Korea the trip for high-level talks a top-secret visit to North Korea top-secret visit to North Korea was talking to North Korea met with North Korean dictator Kim Jong Un met with the North Korean leader
ACTOR	0	JPN\Abe	Abe Mr. Abe Japanese Prime Minister Shinzo Abe a surprisingly brave face the prime minister leader of Japan a friend of Mr. Trump prime minister Prime Minister Shinzo Abe of Japan both leaders a `` truly fine gentleman . " his Japanese counterpart
ACTOR-G	2	KOR\KOR-I	South Korean officials the visiting South Korean envoys officials Mr. Moon 's special envoys an unidentified South Korean official aides to Mr. Moon and Mr. Kim our officials arrogant blockheads a South Korean Duty officers from both sides
ACTOR-G	3	USA\USA-Misc	voters the Pakistani mystery man the American people Trump critics

			three Americans the American public small donors a Pakistan-born information technology worker the public
COUNTRY	9	IRN	Iran Tehran the country the Iranian wolf not been faithful the Shiite nation one of the most exhaustive and robust regimes in the history of nuclear nonproliferation The Iranian regime Iranian deception the regime militant Islam the Tehran government the foremost sponsor of terrorism in the world the inspection regime
EVENT	3	Investigation into RUS meddling	Mueller 's investigation a `` witch hunt . " the special counsel probe that probe investigating what happened in 2016 investigation by Mr. Mueller the Mueller probe investigations into the matter the special counsel 's ongoing Russia probe legislative investigations Special Counsel Robert Mueller 's investigation of Russian meddling Mr. Mueller 's investigation the investigation into Russia 's election meddling investigating contacts between the Trump campaign and Russia the special counsel investigation Mueller 's ongoing criminal investigation the special counsel 's criminal investigation probe Russian interference in the 2016 election

			the special counsel 's Russian meddling and collusion probe the whole Russia investigation special counsel Robert Mueller 's investigation of Russian interference in the 2016 presidential election special counsel Robert Mueller 's investigation into Russian interference in the 2016 election
GROUP	1	USA\Republicans	the chairmen GOP lawmakers The Republican officials the Republican congressmen Republican Congressional intelligence officials The three Republican chairmen three House Republican committee chairmen Conservatives The chairmen of the three House committees
GROUP	5	GBR\GBR-Misc\Demonstrators	thousands of our supporters `` thousands of our supporters " tens of thousands of demonstrators supporters demonstrators 28,000 attendees those opposing President Trump 's visit Mr. Trump 's critics some people who will want to express their views loudly and peacefully
MISC	2	Threats to PRK	seek an agreement in which North Korea promised North Korea `` fire and fury like the world has never seen , "</p> <p>The threats</p> <p>continually referred to Kim as `` Rocket Man , " pushing North Korea to threaten Guam imposed new sanctions against the country Tensions between North Korea and the U.S. missile strikes argued over which country 's nuclear weapons were more powerful must keep applying pressure with an aim that they abandon their nuclear weapons and missiles
MISC	4	FRA-USA warm relationship	together The wonderful friendship

			their close relationship the relationship between the U.S. and France showered him with compliments his warm relationship with Mr. Macron a testament to the enduring friendship that binds our two nations strong and united a testimony to the historic friendship between our two nations their `` partnership . " any family or friendship kissed each other on the cheek . embraced warmly solidarity for the noble causes of liberty and peace even brushed what he said was a piece of dandruff off of Macron 's suit . the long 241-year friendship between the two countries even brushed what he said was dandruff off Mr. Macron 's jacket . a kiss on the cheek from Macron `` constantly grown more solid . "</p> <p>a testament to an enduring friendship that binds our two nations</p> <p>`` a very special relationship , " held Macron 's hands his personal friendship with Macron
OBJECT	1	Comey Memos\Steele dossier	the dossier ` the hookers thing ' nonsense the Steele dossier a vast conspiracy between the Trump campaign and Russia the allegations allegations the salacious dossier detailing Mr. Trump 's ties to Russia allegations about Russian prostitutes a supposed encounter between Mr. Trump and Russian prostitutes consorted with Russian prostitutes the document a dossier the dossier "

			the infamous Steele dossier salacious , unsubstantiated allegations related to personal conduct the salacious dossier produced by a former British spy salacious allegations Trump 's involvement with prostitutes during a 2013 visit in Moscow unsubstantiated allegations the ` Golden Showers thing '
ORGANIZA-TION	6	Supporters\People without borders	organizers the caravan 's organizers which was intended to offer protection to migrants by traveling safer as group and also call attention to the poverty , gang violence and political turmoil in Central America driving families to flee Pueblos Sin Fronteras Pueblo Sin Fronteras , the transnational group that organized the caravan the caravan 's main organizing group , Pueblos Sin Fronteras not the same group as the Washington-based non-profit that provides computer classes to international residents of the District of Columbia caravan organizers the Pueblo Sin Fronteras People Without Borders `` smugglers '' Pueblo Sin Fronteras , a transnational group that organized the caravan